ELEMENTS AND COMPOUNDS

Name: ..... **Date:** .....

Part 1: write each substance in the correct column. Part 2: count the atoms. Part 3: answer in full sentences.

PART 1: ELEMENT OR COMPOUND?

He (helium) · O_2 (oxygen gas) · H_2O (water) · NaCl (salt) · Fe (iron) · CO_2 (carbon dioxide) · N_2 (nitrogen gas) · CH_4 (methane) · Ar (argon) · Cl_2 (chlorine gas) · NH_3 (ammonia) · Cu (copper) · H_2 (hydrogen gas) · MgO (magnesium oxide) · $C_6H_{12}O_6$ (glucose) · Ne (neon)

Element: single atoms	Element: molecules	Compound

PART 2: COUNT THE ATOMS

1. H_2O: _____ hydrogen, _____ oxygen; total _____
2. CO_2: _____ carbon, _____ oxygen; total _____
3. NH_3: _____ nitrogen, _____ hydrogen; total _____
4. $C_6H_{12}O_6$: _____ carbon, _____ hydrogen, _____ oxygen; total _____
5. $Ca(OH)_2$: _____ calcium, _____ oxygen, _____ hydrogen; total _____
6. $2H_2O$ (two molecules of water): _____ hydrogen, _____ oxygen; total _____

PART 3: EXPLAIN

Write a definition of a compound in your own words.

_____

_____

O_2 is a molecule but not a compound. Why?

_____

_____

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ELEMENTS AND COMPOUNDS ANSWER KEY

Name: ..... **Date:** .....

PART 1: ELEMENT OR COMPOUND?

Element: single atoms: He (helium), Fe (iron), Ar (argon), Cu (copper), Ne (neon)

Element: molecules: O_2 (oxygen gas), N_2 (nitrogen gas), Cl_2 (chlorine gas), H_2 (hydrogen gas)

Compound: H_2O (water), NaCl (salt), CO_2 (carbon dioxide), CH_4 (methane), NH_3 (ammonia), MgO (magnesium oxide), $C_6H_{12}O_6$ (glucose)

PART 2: COUNT THE ATOMS

1. 2 H, 1 O; total 3 2. 1 C, 2 O; total 3 3. 1 N, 3 H; total 4 4. 6 C, 12 H, 6 O; total 24 5. 1 Ca, 2 O, 2 H; total 5 6. 4 H, 2 O; total 6

PART 3: EXPLAIN

1. Sample: a substance made of two or more different elements chemically joined together in a fixed ratio. 2. It contains atoms of only one element (oxygen); a compound needs at least two different elements.

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